# ECON 2 Fall 2025 final

Name: _____ SID: _____ · 90 minutes · 60 points · one page of notes, front and back · calculator allowed

## How to use this page

Set a timer for 90 minutes and a blank sheet of paper next to you. Work every part on paper before you open its answer, because the point is to find out what you can do cold. Bring a ruler for the graphs, the same way you will on Dec 15.

Every answer below has three pieces. The first is the full reasoning in plain sentences. The second is a "Full credit" line that says what the official key wrote, which is what a grader looks for. The third is a short note on what the question tests and which unit of this site teaches it.

The question text is copied from the exam with small cleanups: math symbols are written out, and the long cover-page rules are summarized. Answers marked as the official key match the red text in Edwards' Fall 2025 answer key. Anything extra, such as a dollar figure for the real price of a Big Mac, is extra arithmetic that is not in the key, and it is labeled that way.

| Question | Points | Topic | Teaching unit |
|---|---|---|---|
| Free points | 2 | Name and student ID on every page; a fun question | How the final works |
| Q1 (a) to (f) | 12 | Six multiple-choice items, 2 points each | Mixed: see each part |
| Q2 (a) to (e) | 10 | Real price of a Big Mac, automation, bond prices, the price of capital | 06, 03, 13 |
| Q3 (a) to (j) | 24 | Keynesian cross: intercept, equilibrium, two graphs, a shock, the multiplier, offsetting with G | 08, 09 |
| Q4 (a) to (d) | 12 | Soybean exports, the trade war, the farm labor market, winners and losers | 05, 01 |
| **Total** | **60** | | |

## The 2 free points

The cover page gives 1 point for writing your name and student ID at the top of each page, and 1 point for answering a fun question. In Fall 2025 the fun question asked what color your lightsaber would be if you were a Jedi in "Star Wars," and it said there were no wrong answers. Do both in the first minute. Losing a point because one page has no name on it is the most avoidable loss on the exam.

The cover also says the exam is open book and open note with notes limited to one page double-sided, that stand-alone calculators are allowed, that a ruler may be useful, and that you must stop when time is called or lose 10 points. Rule 6 says to keep answers inside the box and ends with "Less is more."

## Question 1: multiple choice (12 points)

The title was "Multiple choices cheer multiple majors? Well, cheers!" The instruction was to circle the one correct answer. Each part is worth 2 points, and there is no partial credit on a circled letter, so read every option before circling.

**2 points** **(a)** A central element of economics is how markets help participants match the level of their marginal benefit to the level of

(i) Marginal utility
(ii) Marginal product
(iii) Marginal cost
(iv) Marginal derivative

**2 points** **(b)** The principle of comparative advantage implies that

(i) I should specialize in doing the task that I do best
(ii) I should specialize in all tasks that I do better than others
(iii) I should specialize in doing the task that is respected most by others
(iv) All of the above

**2 points** **(c)** In economics, the word "capital" means

(i) Productive physical assets like durable equipment and structures
(ii) Financial assets like a bank account or a mutual fund investment
(iii) Accumulated savings that can be invested across international borders
(iv) All of the above

**2 points** **(d)** If the yen-dollar exchange rate falls from ¥160 per U.S. dollar to ¥150 per U.S. dollar, then we say

(i) The yen has appreciated and the dollar has depreciated
(ii) The yen has depreciated and the dollar has appreciated
(iii) Nothing has happened
(iv) We cannot tell based on this information

**2 points** **(e)** Which of these is an important aspect of "social infrastructure," the basic building blocks of a society that allows markets to function well?

(i) The rule of law
(ii) Property rights
(iii) Protection from government
(iv) All of the above

**2 points** **(f)** Compared to perfect competition, a monopolist

(i) Raises price and raises quantity
(ii) Reduces price and raises quantity
(iii) Reduces price and reduces quantity
(iv) Raises price and reduces quantity

## Question 2: short answers (10 points)

The title was "To optimize, set Marginal Short Questions = Marginal Short Answers." Each part is worth 2 points and has a small answer box.

**Setup for (a) and (b):** In 2021, The Economist reported that the average price of a McDonald's Big Mac in the U.S. was $1.60 in 1986 and $5.65 in 2021, or about 3.5 times higher. The consumer price index (CPI, with 1982 to 1984 = 100) was about 110 in 1986 and roughly 271 in 2021, or about 2.5 times higher.

**2 points** **(a)** Has the price of a Big Mac increased, decreased, or stayed the same in inflation-adjusted real terms? That is, if your income increased at the rate of inflation, could you buy fewer, more, or the same numbers of Big Macs today vs. 1986? Explain.

**2 points** **(b)** Is the production and serving of a Big Mac something that could be automated in factories, or is it something that is done by human workers? Discuss the implications for marginal costs over time, given that factory automation has often driven falling costs.

**Setup for (c), (d), and (e):** The present value (PV) of a bond or a claim to capital that pays $F_t$ in year t over 3 years, when the nominal interest rate is i, is

$$PV = F_1 \div (1 + i) + F_2 \div (1 + i)^2 + F_3 \div (1 + i)^3$$

A bond's payouts $F_t$ are fixed. For a unit of capital, $F_t$ is its marginal revenue product of capital, which is marginal revenue times the marginal product of capital: $MRPK(t) = MR(t) \times MPK(t)$.

**2 points** **(c)** Suppose the interest rate i increases. What happens to the price of this bond?

**2 points** **(d)** Suppose i = 0. Express the price of this bond in terms of $F_1$, $F_2$, and $F_3$.

**2 points** **(e)** What happens to the price of a unit of capital when capital's future marginal product falls, for example because of an expected future reduction in labor?

## Question 3: the Keynesian cross (24 points)

The title was "Risks to the U.S. economy in 2025: We Keynesian Cross Ourselves." This is the biggest question on the exam, 40% of the points. Planned aggregate expenditure (PAE) is total planned spending: PAE = C + I + G + NX, where C is consumption, I is investment, G is government purchases, and NX is net exports. Y is real GDP (output). The marginal propensity to consume (mpc) is the share of each extra dollar of income that households spend.

**Setup:** In 2025, U.S. GDP is about $30 trillion. A reasonable estimate of the mpc is 1/3, and the following system models PAE in trillions of dollars:

C = 10 + (1/3)Y  I = 6  G = 5  NX = −1 (net exports are negative $1 trillion)

**2 points** **(a)** Combine the expenditure terms in PAE and find the level of PAE when Y = 0, which is the vertical intercept in the Keynesian cross diagram. Show your work.

**2 points** **(b)** Using your answer to part (a), set PAE = Y and solve to find the short-run equilibrium level of Y, which is the horizontal coordinate of the spot where PAE crosses the 45° line. Show your work.

**4 points** **(c)** Graph the PAE line and the 45-degree line on the axes below, and show the equilibrium level of total output in the short run without any shocks. Label intercepts, etc. (The exam's axes ran from 0 to 35 on both, marked every 5.)

**Setup for (d) to (j):** At the end of 2025, there were concerns about the U.S. economy. Tariffs may raise domestic prices and reduce consumption. Investment in data centers fueled recent GDP growth, but the AI boom could be a bubble that bursts. Immigration enforcement detained and removed immigrant workers, and the net effects remain murky. Suppose these headwinds produce this situation in early 2026:

`C' = 8 + (1/3)Y  I' = 5  G = 5  NX' = 0  PAE' = C' + I' + G + NX'`

**2 points** **(d)** Combine the expenditure terms in PAE' and find the level of PAE' when Y = 0, which is the vertical intercept. Show your work.

**2 points** **(e)** Using your answer to part (d), set PAE' = Y and solve to find the short-run equilibrium level of Y. Show your work.

**4 points** **(f)** On the axes below, graph the original PAE line from part (a), the new PAE' line from part (d), and the 45-degree line. Show equilibria and label intercepts, etc.

**2 points** **(g)** By how much did PAE change between (a) and (d)? Express your answer in trillions of dollars and describe it as an increase, decrease, or no change. Explain fully, making clear what the arithmetic actually means about planned aggregate spending, and describing how it appears in the graph.

**2 points** **(h)** By how much did GDP change between (a) and (d)? Express your answer in trillions of dollars and describe it as an increase, decrease, or no change. Explain fully, making clear what the arithmetic actually means about GDP, and describing how it appears in the graph.

**2 points** **(i)** Are your answers to (g) and (h) the same, or are they different? If they are the same, state as much and explain why. If they are different, identify which one is larger in magnitude, and explain why. Explain fully, referencing aspects of the model.

**2 points (j)** Suppose Congress could pass a temporary spending bill in 2026 that could quickly change G by enough to fully counteract the impacts that you described in parts (d) to (f). How big of a change in G would that be? Would it be a decrease, an increase, or no change in G? Explain fully.

## Question 4: soybeans and the trade war (12 points)

The title was "Oy, Soybeans! So you been exporting?" It mixes the trade graph from Class 16 with the labor market from Class 12.

**4 points (a)** The supply and demand diagram shows the domestic U.S. market for soybeans. Adapt the diagram to show roughly half of the quantity supplied as exports to the world market, meaning the domestic quantity supplied exceeds the domestic quantity demanded, at a world price that compensates suppliers for their marginal costs. Show the world price and the quantity of exports. Briefly explain what you drew.

**Setup for (b):** In spring 2025, the U.S. escalated its trade war with China, which had been the main importer of U.S. soybeans. In May, China responded to higher U.S. tariffs with its own steep tariffs, which practically halted its imports of U.S. soybeans. Assume all U.S. soybean exports had gone to China and no other country will import U.S. soybeans at any price.

**2 points (b)** Producers plant soybeans in the spring and harvest in the fall. If producers planted before the trade war escalation and cannot change that quantity, and if soybeans cannot be stored or destroyed, how much must the price adjust, if at all? Explain and show the price level after the trade war escalation on the graph in part (a).

**Setup for (c):** Consider the labor market for workers in U.S. soybean production. The labor supply curve reflects workers' marginal cost of supplying their labor. The labor demand curve is the marginal revenue

product of labor, MRPL = MR × MPL, where MR is the marginal revenue from selling one more bushel and MPL (the marginal product of labor) is how many extra bushels one more worker produces.

**4 points (c)** Illustrate the situation in the labor market before and after the trade war escalation as we have described it up to now. Show any and all demand and supply curves and equilibrium wages and employment. Briefly explain.

**Setup for (d):** In late October 2025, the U.S. and China reached a truce in which China agreed to buy more U.S. soybeans. By early December, the U.S. announced a one-time government payment of $12 billion to U.S. soybean farmers, and China reported that its trade surplus reached a record high.

**2 points (d)** What do you think these three late-2025 events reveal about the winners and losers of this round of the trade war, and given what you illustrated earlier, why? Recall that imports from China are usually durable manufactured goods, not perishable food.

## What this exam tells you about Dec 15

- Thirty-six of the 60 points (60%) came from two long stories: a Keynesian cross with numbers and a trade problem that ended in the labor market. Expect the same shape again.
- Three of the six multiple-choice items (a, b, f) were midterm material: MB = MC, comparative advantage, and monopoly. One short answer (Q2 b) was a marginal cost argument. That is why Midterm material refresher exists.
- The four graph parts, Q3(c), Q3(f), Q4(a), and Q4(c), were worth 4 points each, 16 points total. Every one needed labeled curves, labeled intercepts or prices, and dotted lines from each equilibrium to both axes.
- The key's answers are short: a direction word (increased, decreased, falls), a number with units, and one sentence of model reasoning with the right term (marginal revenue, marginal propensity to consume, multiplier).
- The Keynesian cross question walked you step by step. If you get stuck on (a), use the setup equations to keep going; every later part builds on the same intercept and slope.